

Sushanth Buddhala

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San Ramon, California

PROFILE

Data Analytics Engineering graduate with hands-on internship experience in infrastructure automation and AI-powered retrieval systems, skilled in Python, SQL, Power BI, and computer networking fundamentals. Focused on prompt engineering, data validation, and translating complex datasets into actionable insights.

WORK EXPERIENCE

Software Engineering Intern- Apposite Technologies LLC

Jul 2026 - Current

- Developing a CLI automation tool to provision network emulator virtual environments across three deployment targets: QEMU/KVM, VMware ESXi, and AWS.
- Implementing infrastructure provisioning logic in Python with OpenTofu/Terraform, and configuring VM initialization using cloud-init and JSON-based config files.
- Collaborating directly with Apposite's engineering team through kickoff and recurring print check-ins to scope provisioning workflows and requirements.

PROJECTS

Compass- Agentic Multi-Agent Retail Analytics System

Feb 2026 - May 2026

- Architected a 5-agent system (Planner, Data, Product, Supply Chain, Synthesizer) in Python using Gemini API function calling and Pydantic orchestrator on DuckDB, built with Cursor AI assistance.
- Built a column synonym normalization layer, hybrid Flash/Pro fallback strategy, and hand-crafted system prompts for all 5 agents through iterative prompt engineering; deployed a Streamlit chat UI with inline Plotly charts and reasoning trace visualization.
- Designed a 40-case integration eval harness with 4-dimensional scoring (correctness, routing, SQL quality, overall) using Pytest, reaching 95% completion and 95.2% correctness across 3 iterative bug-fix rounds.

SARRDI - Dam Infrastructure Risk Analytics (Capestone Project)

Jan 2026 - May 2026

- Engineered an ETL pipeline for Virginia DCR, integrating inspection, flood, and social vulnerability data across 2,712 dams via fuzzy matching and watershed-level joins into a unified risk dataset.
- Built a 5-dimension weighted risk-scoring model and Power BI/DAX dashboard under GMU's STAR-TIDES Program, flagging 170 dams as critical priority.
- Delivered field-ready decision support to DCR with zero change requests, supporting statewide dam-safety prioritization.

TECHNICAL SKILLS

Languages & OS: Python, SQL, DAX, Linux, Windows

Frameworks & Libraries: Pydantic, Streamlit, FastAPI, Pytest, Pandas, NumPy, Plotly, Scikit-learn, Tenacity

Databases & Platforms: Microsoft Excel, DuckDB, SQLite, AWS, Cursor AI, Snowflake, Power BI, Databricks, Microsoft Power Query

Data & ML: Data Analysis, Data Modeling, Data Warehousing, Data Pipelines, ETL pipelines, Statistical Analysis, Machine Learning, Unsupervised learning

AI & LLM: Gemini API, OpenAI API, Function Calling, Prompt Engineering, Multi-Agent Systems, Workflow Orchestration, Model Evaluation, RAG Pipelines, Generative AI

EDUCATION

George Mason University

M.S. in Data Analytics Engineering

- **Relevant Coursework:** Analytics & Big Data, Intro to Analytics & Modeling, Principles of Data Management & Mining, Visualization for Analytics, Big Data Essentials, Data Mining for Business Analytics, Interpretable Machine Learning, Data Analytics Project

Gokaraju Rangaraju Institute of Engineering and Technology

B.S. in Computer Science Engineering

- **Relevant Coursework:** Linear Algebra Differential Calculus, Programming for Problem Solving, Differential Equation and Vector Calculus, Computer Networks, Database Management System, Python Programming, Artificial Intelligence, Data Analytics with Python, Machine Learning, Big Data Analytics, Cloud Computing, Data Warehousing and Data Mining

PROFESSIONAL CERTIFICATES

• IBM Data Analyst Professional Certificate by Coursera • AWS Academy Graduate - Cloud Foundation • Generative AI for Data Analytics by Coursera